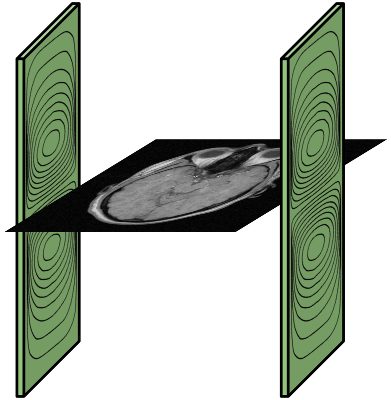
Optimization of Gradient Coils in MRI

Daniel Abraham and Munir Bshara

Outline

MRI & Gradients



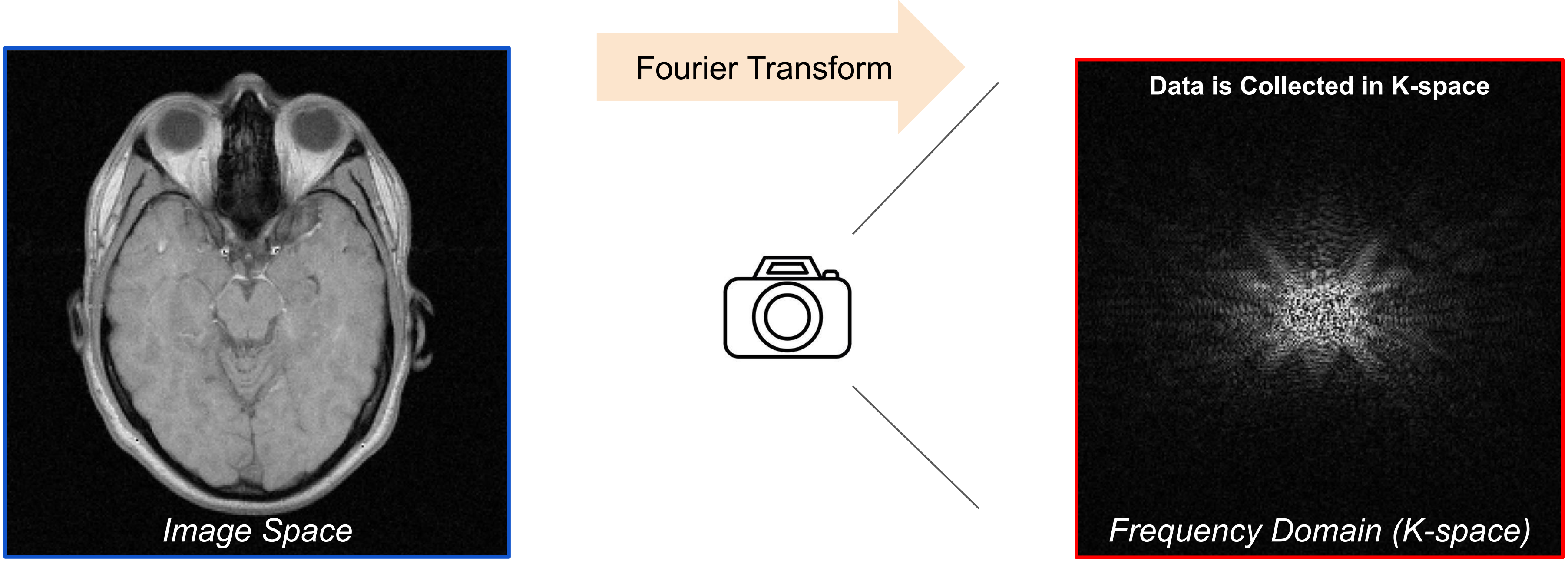
Gradient Design and EM Modeling

Optimization & Computational Methods

$$
\begin{aligned}
\min_{\theta,\mathbf{x},G_{\min},L_{\max},E_{\max}} \quad & -\lambda_G G_{\min} + \lambda_L L_{\max} + \lambda_E E_{\max} \\
\text{s.t.} \quad & \\
& \mathbf{e}_d^T \mathbf{G}_\theta(\mathbf{r})\mathbf{x} \geq G_{\min}, \forall \mathbf{r} \in \Omega_{\mathrm{dsv}} \\
& ||\mathbf{E}_\theta(\mathbf{r})\mathbf{x}||_2 \leq E_{\max}, \forall \mathbf{r} \in \Omega_{\mathrm{pns}} \\
& \frac{1}{2} L_{\theta,n} x_n^2 \leq L_{\max}, \forall n \in \{1, \cdots, N\}.
\end{aligned}
$$

Simulation Results

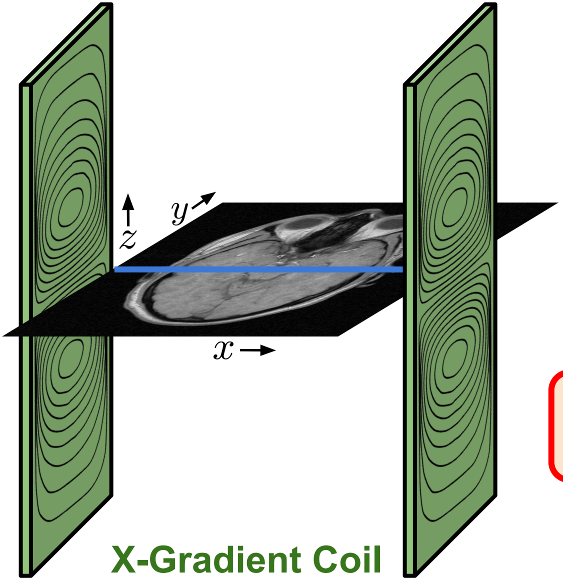
MRI Background: K-space Model

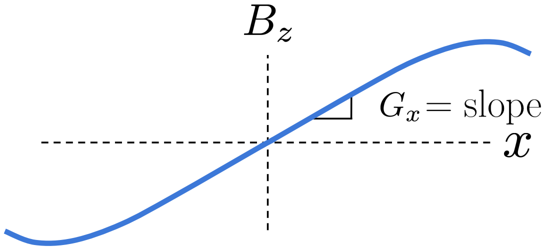


Data acquisition speed is dependent
on how quickly we can **traverse k-space**

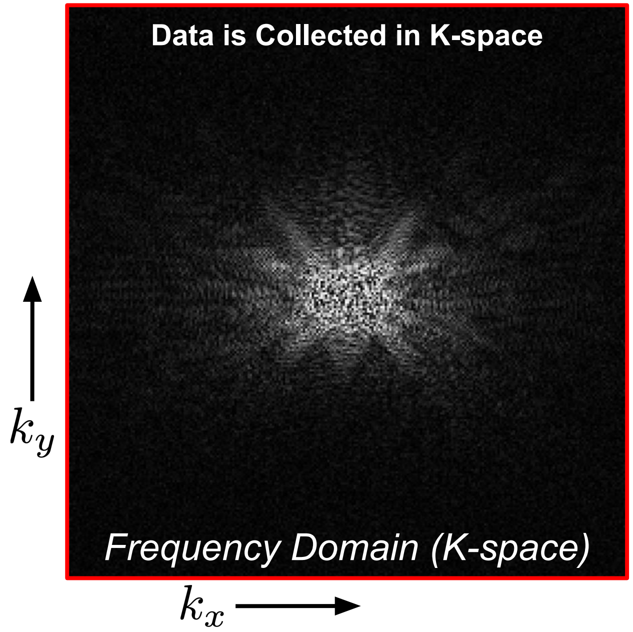
MRI Background: Gradient Coil

$$k_x(t) \propto \int_t G_x(\tau)d\tau \qquad k_y(t) \propto \int_t G_y(\tau)d\tau$$





Strong Gradient → Fast Imaging

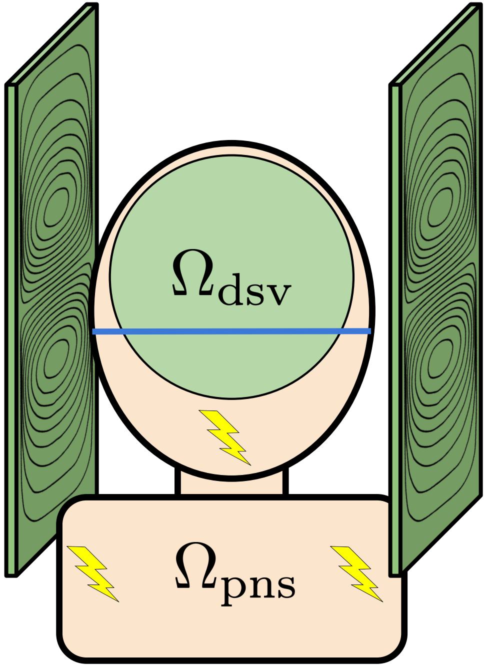


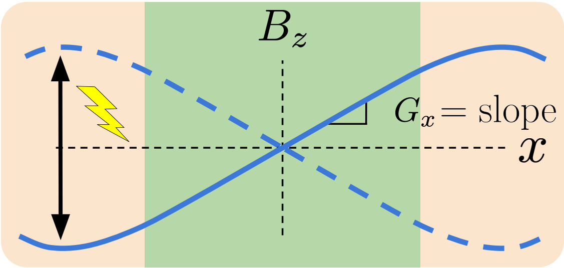
Data acquisition speed is dependent
on how quickly we can **traverse k-space**

K-space traversal is controlled by **gradients fields**,
which are linearly varying fields pointing in the z-direction

MRI Background: Gradient Coil *Limitations*

$$k_x(t) \propto \int_t G_x(\tau)d\tau \qquad k_y(t) \propto \int_t G_y(\tau)d\tau$$





We want a **strong** gradient field over the **design spherical volume (DSV) region** for **fast imaging**

$$-\frac{\partial}{\partial t}G_x(t)\cdot X_{\max} = \nabla \times E$$

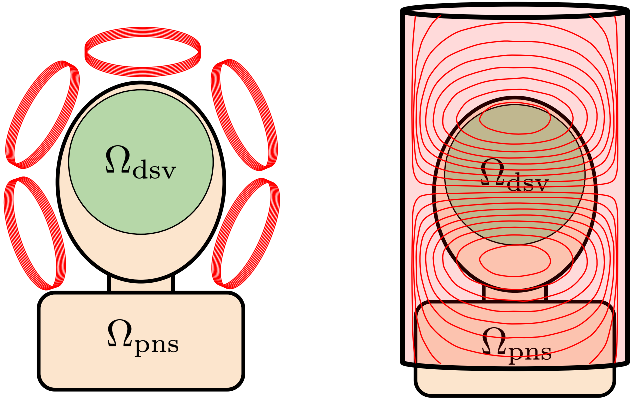
But rapidly switching gradients induce **stray electric fields**, leading to **peripheral nerve stimulation (PNS)**

Thus gradient coils must **maximize the linear field** while **minimizing stray electric fields** to enable high gradient strength and rapid gradient switching applications

Outline

MRI & Gradients

Gradient Design and EM Modeling



Optimization & Computational Methods

$$\min_{\boldsymbol{\theta},\mathbf{x},G_{\min},L_{\max},E_{\max}} -\lambda_G G_{\min} + \lambda_L L_{\max} + \lambda_E E_{\max}$$

$$\text{s.t.}$$

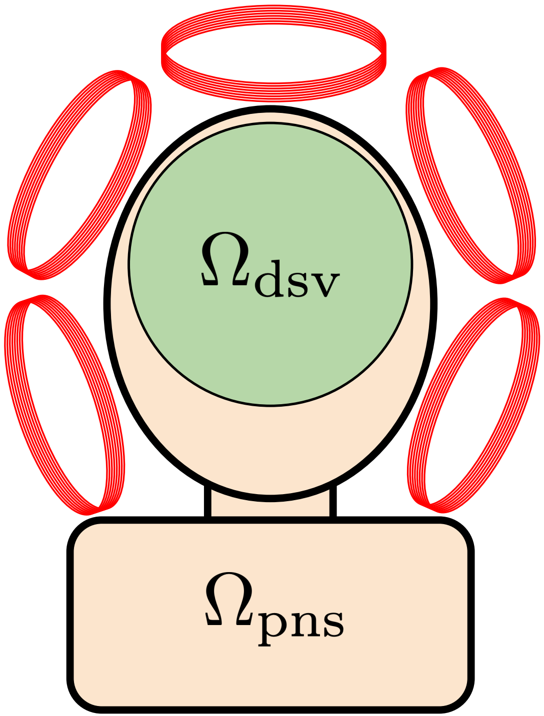
$$\mathbf{e}_d^T \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x} \geq G_{\min}, \forall \mathbf{r} \in \Omega_{\mathrm{dsv}}$$

$$||\mathbf{E}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x}||_2 \leq E_{\max}, \forall \mathbf{r} \in \Omega_{\mathrm{pns}}$$

$$\frac{1}{2} L_{\boldsymbol{\theta},n} x_n^2 \leq L_{\max}, \forall n \in \{1, \cdots, N\}.$$

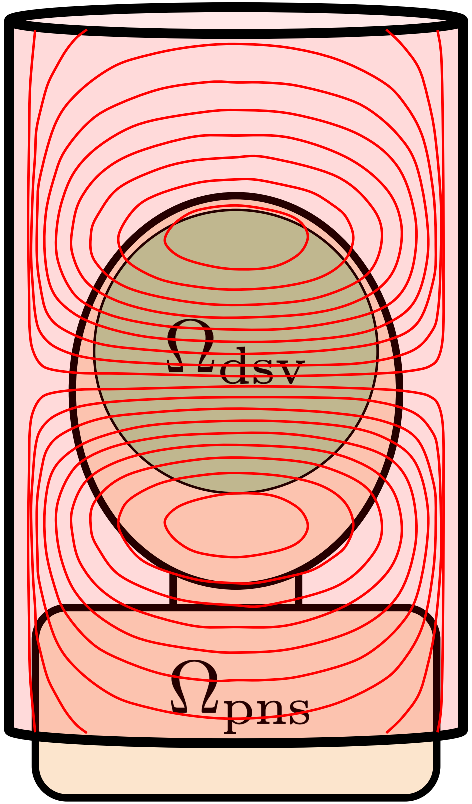
Simulation Results

Two Common Gradient Coil Approaches



Target gradient field is linearly synthesized via fields from multiple **independent coil elements**

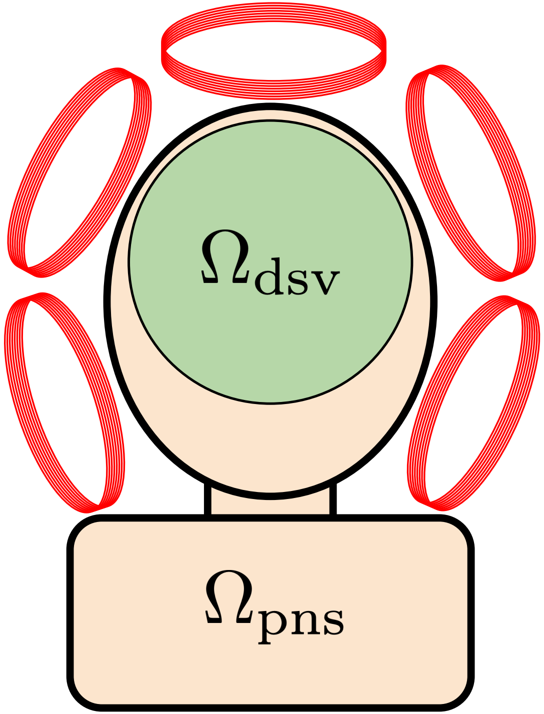
Target gradient field is built through a single carefully designed connected **wire pattern**



Matrix Gradient Design

Stream Function Design

Two Gradient Coil Approaches



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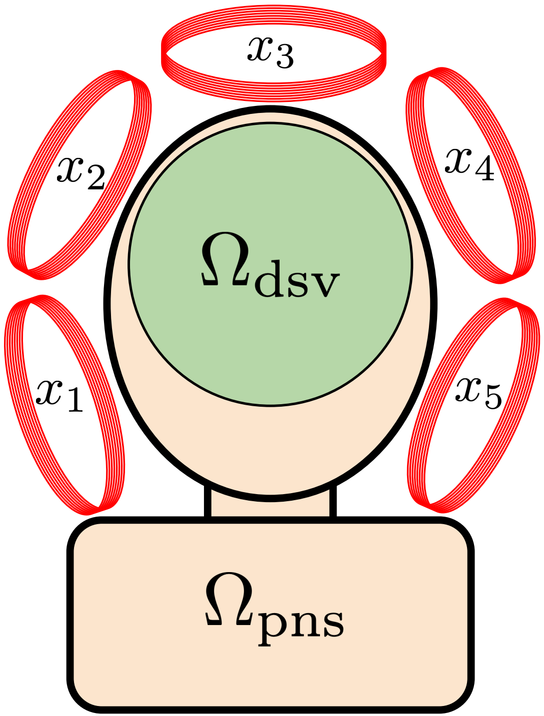
Matrix Gradient Design

Target gradient field is built through a single carefully designed connected **wire pattern**

Stream Function Design

The Matrix Gradient Coil



Matrix Gradient Design

x_k – Ampere-turn value for coil k

$\mathbf{x} = (x_1, \cdots, x_5)$ parameterizes the design

Analytic Solutions for Circular Coils

Magneto-Static Integral Equations

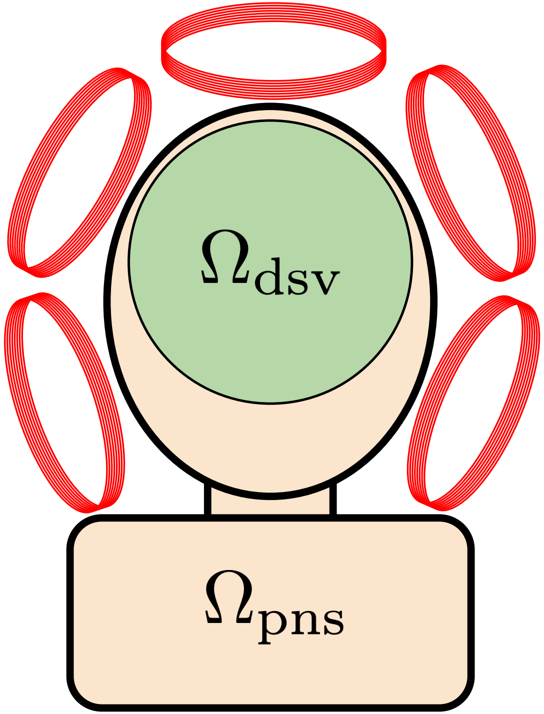
$$\frac{\partial}{\partial \mathbf{r}} B_k^{(z)}(\mathbf{r}_{\text{dsv}}) \qquad \mathbf{E}_k(\mathbf{r}_{\text{pns}}) \qquad L_k$$

$\mathbf{Gx}$ — *Gradient Field*

$\mathbf{Ex}$ — *Electric Field*

$L_k x_k^2$ — *Magnetic Energy*

Two Gradient Coil Approaches



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Target gradient field is built through a single carefully designed connected **wire pattern**

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Two Gradient Coil Approaches

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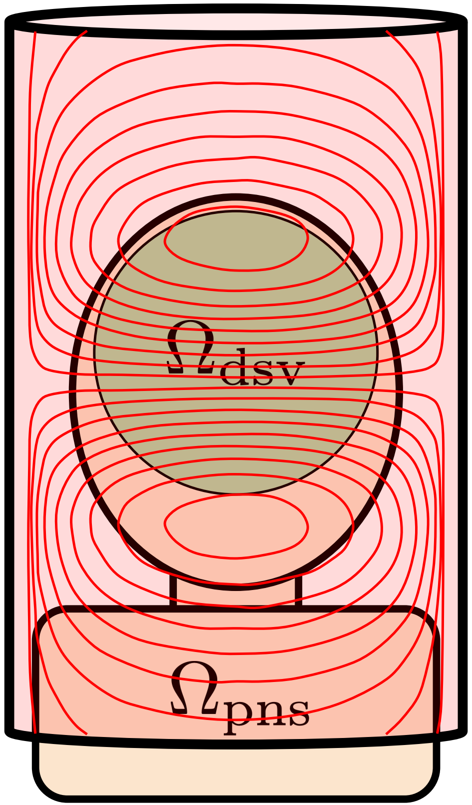


Stream Function Design

Stream Function Setup

We want to model a continuously varying surface current density $\mathbf{J}(\theta, z)$ while keeping it divergence free

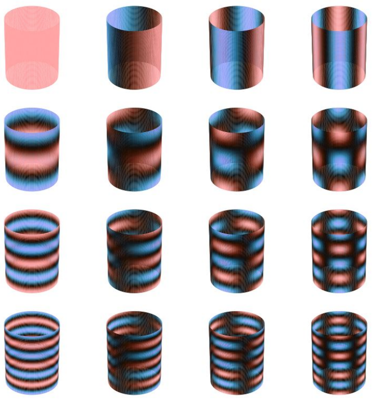
$$\nabla_{\mathrm{S}} \cdot \mathbf{J} = 0$$

We can enforce a divergence free current density using a stream function $\phi(\theta, z)$ construction

$$\mathbf{J} = \nabla_S \phi \times \hat{\mathbf{n}}$$

The stream function is described using **smooth basis functions**

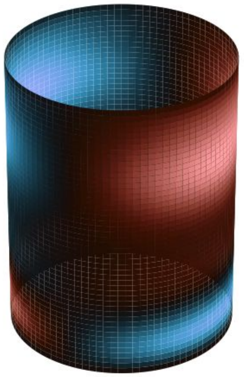
$$\phi(\theta, z) = \sum_{m,n} \underline{B_{m,n}(\theta, z)}\, \phi_{m,n}$$



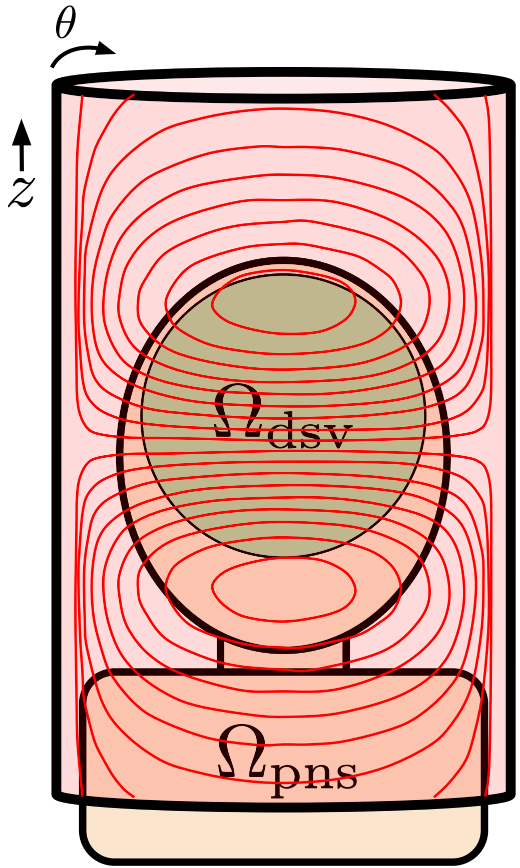
Stream Basis Functions

$$\mathbf{x} = \begin{bmatrix} \phi_{1,1} \\ \vdots \\ \phi_{M,N} \end{bmatrix}$$

Stream Basis Coefficients

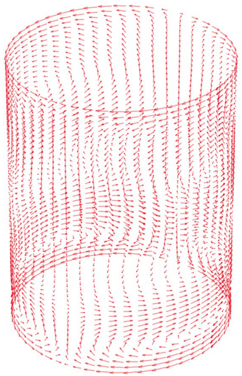


X-Gradient Example $\phi(\theta, z)$



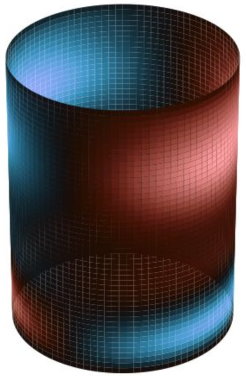
Stream Function Design

Stream Function Pipeline



Current Density $\mathbf{J}(\theta, z)$

$\mathbf{J} = \nabla_S \phi \times \hat{\mathbf{n}}$

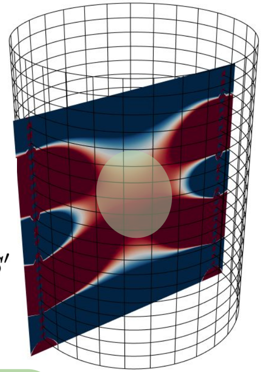


Stream Function $\phi(\theta, z)$

Marching Squares Contouring Algorithm

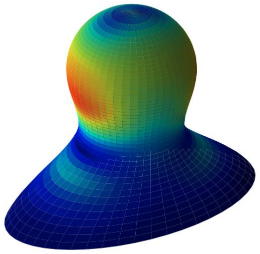
Discrete Winding Pattern

$$\frac{\mu_0}{4\pi}\int_{S'} \frac{\mathbf{M}\mathbf{J}_n(\mathbf{r}')}{||\mathbf{r}-\mathbf{r}'||^3} dS'$$
$$\frac{-3\mu_0}{4\pi}\int_{S'} \frac{((\mathbf{r}-\mathbf{r}')^T\mathbf{M}\mathbf{J}_n(\mathbf{r}'))(\mathbf{r}-\mathbf{r}')}{||\mathbf{r}-\mathbf{r}'||^5} dS'$$



$\mathbf{Gx}$

Gradient Field



$$\frac{\mu_0}{4\pi}\int_{S'} \frac{\mathbf{J}_n(\mathbf{r}')}{||\mathbf{r}-\mathbf{r}'||} dS'$$

$\mathbf{Ex}$

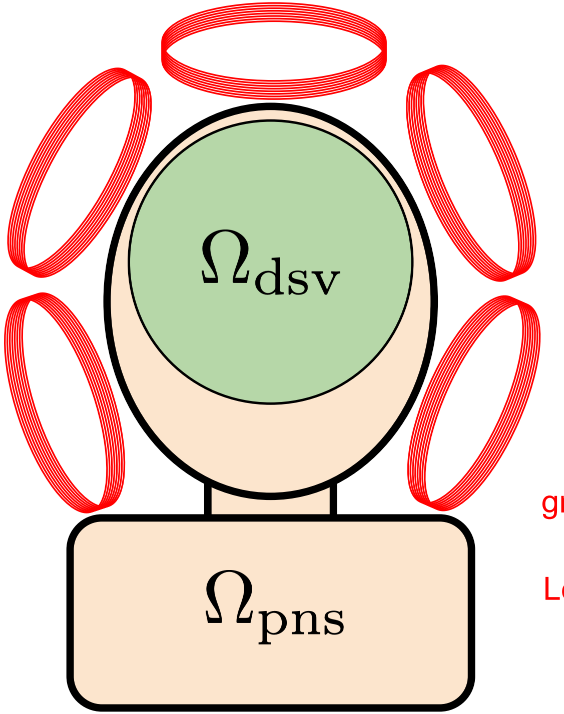
Electric Field

$$\frac{\mu_0}{8\pi}\int_{S}\int_{S'} \frac{\mathbf{J}_m(\mathbf{r}) \cdot \mathbf{J}_n(\mathbf{r}')}{||\mathbf{r}-\mathbf{r}'||} dS' dS$$

$\mathbf{x}^T\mathbf{L}\mathbf{x}$

Magnetic Energy

Stream vs Matrix Gradient



Matrix Gradient Design

Software controlled gradient shape

Lower per channel inductance and power

Simple forward model

Need for independent gradient power amplifiers

Less degrees of freedom

Stream Function Design

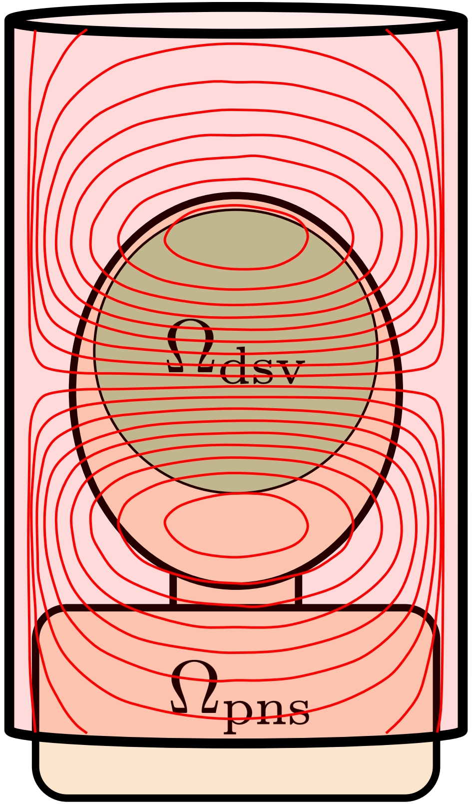
Fixed single gradient axis and shape

Higher inductance and power

Complex forward model

Driven with a single gradient power amplifier

More degrees of freedom



Accurate Computation of Electric Fields

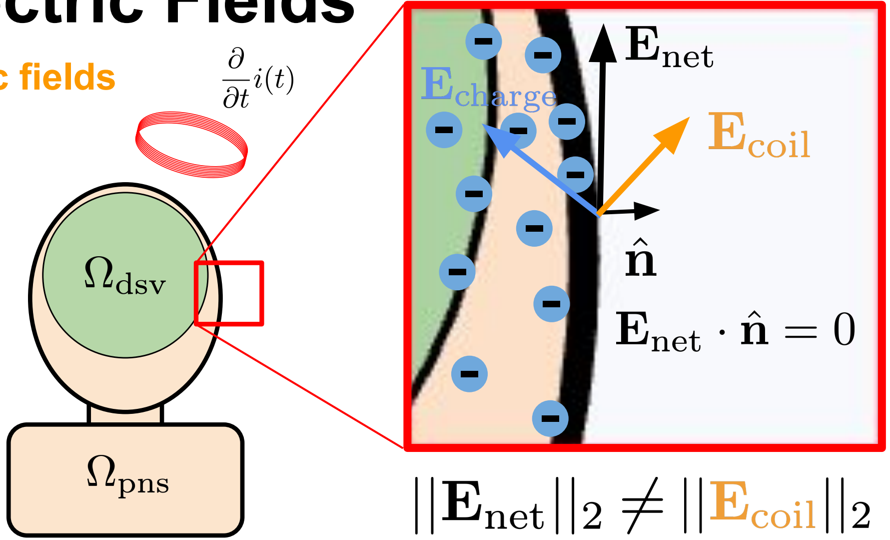
Changes in **current through the coil** will induce **electric fields**

$$\mathbf{E}_{\text{net}} = \mathbf{E}_{\text{charge}} + \mathbf{E}_{\text{coil}}$$

The **net electric field** is a combination of accumulated **surface charges** and the **gradient coil electric field**

$$\mathbf{J}_{\text{net}} \cdot \hat{\mathbf{n}} = 0 \implies (\sigma \mathbf{E}_{\text{net}}) \cdot \hat{\mathbf{n}} = 0$$

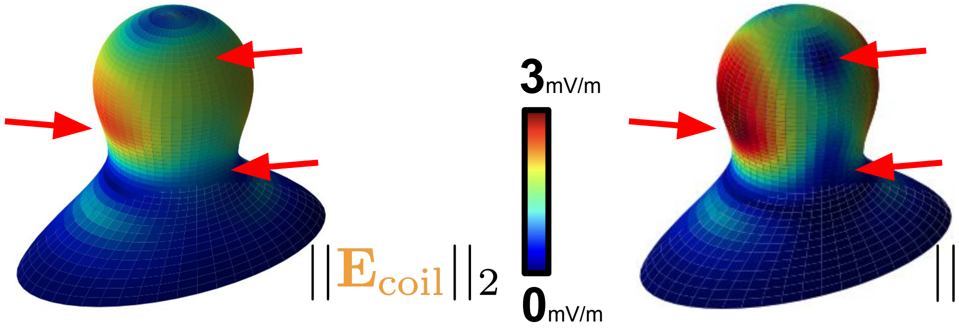
The **surface charges** distribute to cancel normal currents, since charges can't leave the surface. Hence, the **net normal electric field is zero**



$$||\mathbf{E}_{\text{net}}||_2 \neq ||\mathbf{E}_{\text{coil}}||_2$$

Using the boundary condition, we can estimate the **charge distribution** to compute the **net electric field**

Without surface charge model



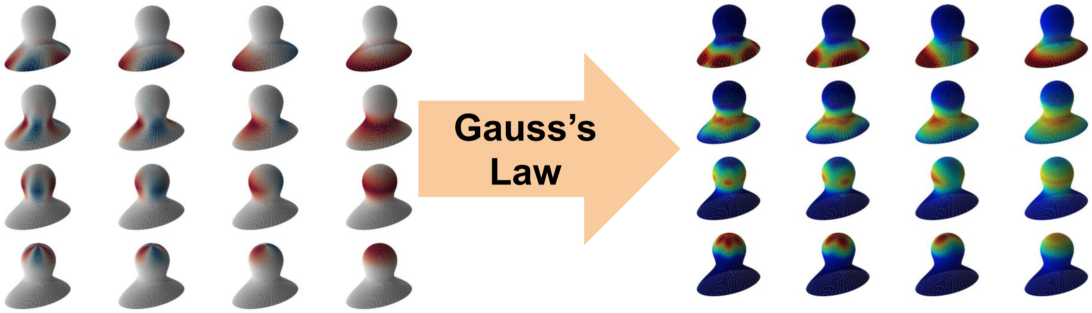
With surface charge model

Roemer, Peter B., et al. "Electric field calculation and peripheral nerve stimulation prediction for head and body gradient coils."

Accurate Computation of Electric Fields

*Surface **charge** basis functions*

*Surface **Efield** basis functions*



$\mathbf{E}_{\text{charge}}$

surface charge coefficients

gradient design coefficients

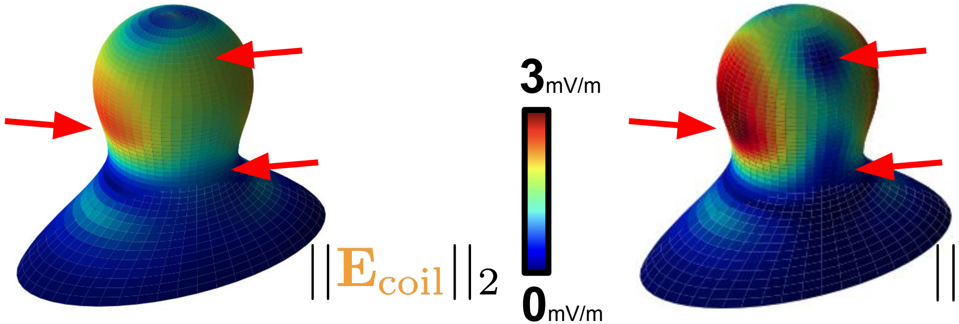
$$\mathbf{E}_{\text{net}} = \mathbf{E}_{\text{charge}}\mathbf{c} + \mathbf{E}_{\text{coil}}\mathbf{x}$$

$$\mathbf{E}_{\text{net}} \cdot \hat{\mathbf{n}} = 0$$ ***for all surface points***

$$\mathbf{c}^{\star} \leftarrow \text{least-squares}$$

Using the boundary condition, we can estimate the charge distribution to compute the **net electric field**

Without surface charge model



$||\mathbf{E}_{\text{coil}}||_2$

$||\mathbf{E}_{\text{net}}||_2$

With surface charge model

Roemer, Peter B., et al. "Electric field calculation and peripheral nerve stimulation prediction for head and body gradient coils."

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$$\min_{\boldsymbol{\theta},\mathbf{x},G_{\min},L_{\max},E_{\max}} -\lambda_G G_{\min} + \lambda_L L_{\max} + \lambda_E E_{\max}$$

$$\text{s.t.}$$

$$\mathbf{e}_d^T \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x} \geq G_{\min}, \forall \mathbf{r} \in \Omega_{\mathrm{dsv}}$$

$$||\mathbf{E}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x}||_2 \leq E_{\max}, \forall \mathbf{r} \in \Omega_{\mathrm{pns}}$$

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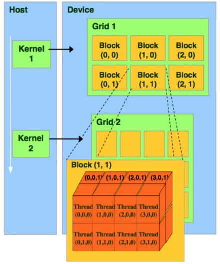
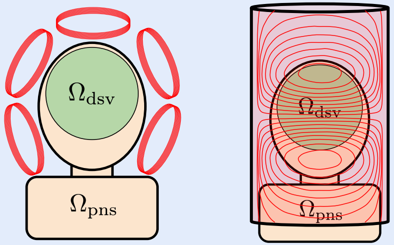
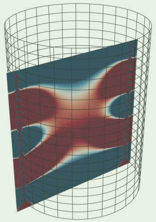


Simulation Results

Optimization Formulation



$\mathbf{x}$ → **coil parameter**

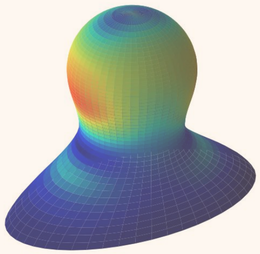


Strong Gradient → Fast Imaging

Create a coil pattern which generates a strong gradient field at low power while mitigating PNS.

$$\frac{\mu_0}{8\pi}\int_S\int_{S'}\frac{\mathbf{J}_m(\mathbf{r})\cdot\mathbf{J}_n(\mathbf{r}')}{||\mathbf{r}-\mathbf{r}'||}dS'dS$$

High Inductance → Slower Imaging



Strong E-Field → Nerve Stimulation

Optimization Formulation

Create a coil pattern which generates a strong gradient field
at low power while mitigating PNS.

$$\min_{\boldsymbol{\theta},\mathbf{x},G_{\min},L_{\max},E_{\max}} -\lambda_G G_{\min} + \lambda_L L_{\max} + \lambda_E E_{\max}$$

s.t.

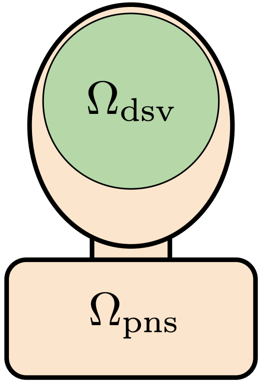
$$\mathbf{e}_d^T \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x} \geq G_{\min}, \forall \mathbf{r} \in \Omega_{\text{dsv}}$$

$$||\mathbf{E}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x}||_2 \leq E_{\max}, \forall \mathbf{r} \in \Omega_{\text{pns}}$$

$$\frac{1}{2} L_{\boldsymbol{\theta},n} x_n^2 \leq L_{\max}, \forall n \in \{1, \cdots, N\}.$$

$\Omega_{\text{dsv}} : \{r_i\}_{i=1}^{N_G}$

$\Omega_{\text{pns}} : \{r_j\}_{j=1}^{N_E}$



How can we limit the search space for the coil patterns to ensure tractability?

Optimization Formulation - Basis Decomposition

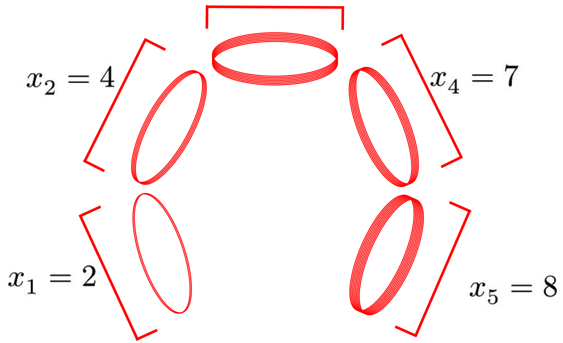
To optimize we need some discrete representation of the parameter space!

$\mathbf{x}, \boldsymbol{\theta}$

Matrix Gradient

Let θ describe the coil geometry, position, eccentricity, etc.

Let $\mathbf{x}$ describe the number of ampere-loops in each coil.



Reminder: $\mathbf{J}(\mathbf{r})dS \rightarrow Idl$

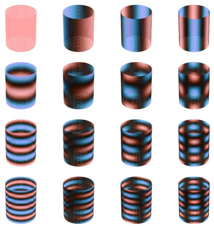
This is the equivalence of Matrix and Stream approaches.

Stream Function

Let θ describe the surface where charge density is being solved for.

Let $\mathbf{x}$ describe the coefficients of the basis functions describing the charge density on the surface.

$$\phi(\theta, z) = \sum_{mn} x_{m,n}\phi_{m,n}(\theta, z) \rightarrow \mathbf{J}(\mathbf{r}) = \sum_{mn} x_{m,n}\mathbf{J}_n(\mathbf{r})$$



Stream Basis Functions

Optimization Formulation - Basis Decomposition

These types of basis decompositions have the super power of being linear in the KPI's!

$$\frac{\mu_0}{4\pi}\int_{S'}\left[\frac{[\mathbf{J}(\mathbf{r})]_\times}{\|\mathbf{r}'-\mathbf{r}\|^3}-3\frac{(\mathbf{J}(\mathbf{r})\times(\mathbf{r}'-\mathbf{r}))\otimes(\mathbf{r}'-\mathbf{r})}{\|\mathbf{r}'-\mathbf{r}\|^5}\right]dS'$$

$$\frac{\mu_0}{4\pi}\int_{S'}\frac{\mathbf{J}(\mathbf{r}')}{\|\mathbf{r}'-\mathbf{r}\|}dS'$$

$$\frac{\mu_0}{8\pi}\int_S\int_{S'}\frac{\mathbf{J}(\mathbf{r}')\cdot\mathbf{J}(\mathbf{r})}{\|\mathbf{r}'-\mathbf{r}\|}dSdS'$$

$$\mathbf{Gx}$$

Gradient Field

$$\mathbf{G}(\mathbf{r})=\sum_n x_n\mathbf{G}_n(\mathbf{r})$$

$$\mathbf{Ex}$$

Electric Field

$$\mathbf{E}(\mathbf{r})=\sum_n x_n\mathbf{E}_n(\mathbf{r})$$

$$\mathbf{x}^T\mathbf{Lx}$$

Magnetic Energy

$$\mathbf{W}=\sum_n\sum_m x_n x_m\mathbf{L}_{m,n}$$

LINEAR IN $\mathbf{J}$

LINEAR IN $\mathbf{J}$

QUADRATIC IN $\mathbf{J}$

Discrete Integral Computation - Quadrature

Create a coil pattern which generates a strong gradient field at low power while mitigating PNS.

$$\min_{\boldsymbol{\theta},\mathbf{x},G_{\min},L_{\max},E_{\max}} -\lambda_G G_{\min} + \lambda_L L_{\max} + \lambda_E E_{\max}$$

s.t.

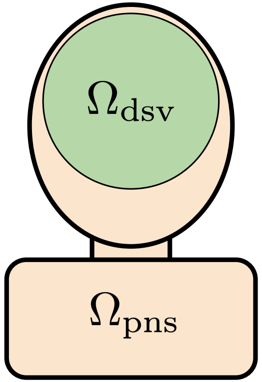
$$\mathbf{e}_d^T \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x} \geq G_{\min}, \forall \mathbf{r} \in \Omega_{\mathrm{dsv}}$$

$$||\mathbf{E}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x}||_2 \leq E_{\max}, \forall \mathbf{r} \in \Omega_{\mathrm{pns}}$$

$$\frac{1}{2} L_{\boldsymbol{\theta},n} x_n^2 \leq L_{\max}, \forall n \in \{1, \cdots, N\}.$$

$\Omega_{\mathrm{dsv}} : \{r_i\}_{i=1}^{N_G}$

$\Omega_{\mathrm{pns}} : \{r_j\}_{j=1}^{N_E}$



How do we form these KPI matrices?

Discrete Integral Computation - Quadrature

Continuous

$$\mathbf{G}(\mathbf{r}) = \nabla \mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{S'} \left[\frac{[\mathbf{J}(\mathbf{r})]_\times}{\|\mathbf{r}' - \mathbf{r}\|^3} - 3 \frac{(\mathbf{J}(\mathbf{r}) \times (\mathbf{r}' - \mathbf{r})) \otimes (\mathbf{r}' - \mathbf{r})}{\|\mathbf{r}' - \mathbf{r}\|^5} \right] dS'$$

$$\mathbf{E}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{S'} \frac{\mathbf{J}(\mathbf{r}')}{\|\mathbf{r}' - \mathbf{r}\|} dS'$$

$$\mathbf{W} = \frac{\mu_0}{8\pi} \int_S \int_{S'} \frac{\mathbf{J}(\mathbf{r}') \cdot \mathbf{J}(\mathbf{r})}{\|\mathbf{r}' - \mathbf{r}\|} dS dS'$$

Convert $\int \rightarrow \sum$ and $dS \rightarrow \Delta S$

Discrete

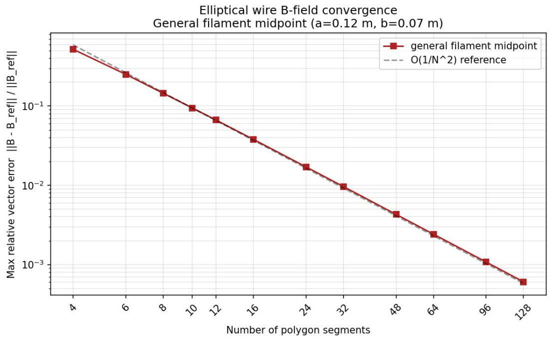
$$\mathbf{G}_n(\mathbf{r}_i) \approx \frac{\mu_0}{4\pi} \sum_{j=1}^{N} \left[\frac{[\mathbf{J}_n^j]_\times}{\|\mathbf{r}_j - \mathbf{r}_i\|^3} - 3 \frac{(\mathbf{J}_n^j \times (\mathbf{r}_j - \mathbf{r}_i)) \otimes (\mathbf{r}_j - \mathbf{r}_i)}{\|\mathbf{r}_j - \mathbf{r}_i\|^5} \right] \Delta S_j \qquad \mathbf{G}(\mathbf{r}) = \sum_n x_n \mathbf{G}_n(\mathbf{r})$$

$$\mathbf{E}_n(\mathbf{r}_i) \approx \frac{\mu_0}{4\pi} \sum_{j=1}^{N} \frac{\mathbf{J}_n^j}{\|\mathbf{r}_j - \mathbf{r}_i\|} \Delta S_j \qquad \mathbf{E}(\mathbf{r}) = \sum_n x_n \mathbf{E}_n(\mathbf{r})$$

$$\mathbf{L}_{m,n} \approx \frac{\mu_0}{8\pi} \sum_{i=1}^{M} \sum_{j=1}^{N} \frac{\mathbf{J}_n^j \cdot \mathbf{J}_m^i}{\|\mathbf{r}_j - \mathbf{r}_i\|} \Delta S_i \Delta S_j \qquad \mathbf{W} = \sum_n \sum_m x_n x_m \mathbf{L}_{m,n}$$

Discrete Integral Computation - Quadrature

If we want panel based quadrature, like investigated in HW1:



Left $\rightarrow \mathbf{J}^j = \mathbf{J}(\mathbf{r}_j)$, Right $\rightarrow \mathbf{J}^j = \mathbf{J}(\mathbf{r}_{j+1})$, Midpoint$\rightarrow \mathbf{J}^j = \mathbf{J}\left(\frac{\mathbf{r}_j+\mathbf{r}_{j+1}}{2}\right)$

Works for both Matrix and Stream approaches.
All of these integrals contain singularities defined by **closeness in measurement point and coil point**. Is this a problem?

Discrete

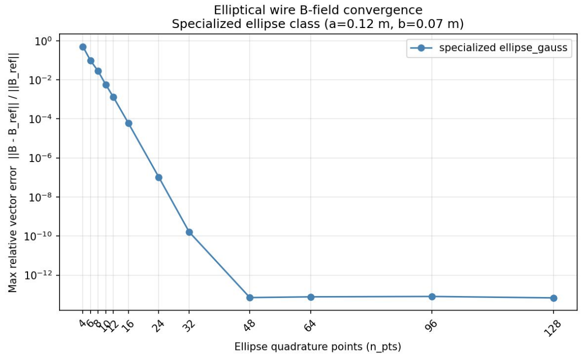
$$\mathbf{G}_n(\mathbf{r}_i) \approx \frac{\mu_0}{4\pi}\sum_{j=1}^{N}\left[\frac{[\mathbf{J}_n^j]_\times}{\|\mathbf{r}_j-\mathbf{r}_i\|^3} - 3\frac{(\mathbf{J}_n^j \times (\mathbf{r}_j-\mathbf{r}_i))\otimes(\mathbf{r}_j-\mathbf{r}_i)}{\|\mathbf{r}_j-\mathbf{r}_i\|^5}\right]\Delta S_j \qquad \mathbf{G}(\mathbf{r}) = \sum_n x_n \mathbf{G}_n(\mathbf{r})$$

$$\mathbf{E}_n(\mathbf{r}_i) \approx \frac{\mu_0}{4\pi}\sum_{j=1}^{N}\frac{\mathbf{J}_n^j}{\|\mathbf{r}_j-\mathbf{r}_i\|}\Delta S_j \qquad \mathbf{E}(\mathbf{r}) = \sum_n x_n \mathbf{E}_n(\mathbf{r})$$

$$\mathbf{L}_{m,n} \approx \frac{\mu_0}{8\pi}\sum_{i=1}^{M}\sum_{j=1}^{N}\frac{\mathbf{J}_n^j \cdot \mathbf{J}_m^i}{\|\mathbf{r}_j-\mathbf{r}_i\|}\Delta S_i \Delta S_j \qquad \mathbf{W} = \sum_n\sum_m x_n x_m \mathbf{L}_{m,n}$$

Discrete Integral Computation - Quadrature

If we are restricted to elliptical loops of wires i.e. Matrix Approach



Parameterize ellipse: $c_n(t), \quad t \in [a, b]$

Approximate Integral: $\oint_{C_n} f(r)dl \approx \frac{b-a}{2}\sum_{q=1}^{Q} w_q f(c_n(t_q))\|c_n(t_q)\|$

Q_n = number of Gauss-Legendre quadrature points on loop n

w_q : corresponding quadrature weights

Reminder: $\mathbf{J}(\mathbf{r})dS \rightarrow Idl$

Discrete

$$G_{n(r_i)} \approx \frac{\mu_0}{4\pi} * \frac{b_n - a_n}{2}\sum_{q=1}^{Q_n} w_q \left[\frac{[c_{n'}(t_q)]_\times}{\left|c_{n(t_q)} - r_i\right|^3} - 3\frac{\left(c_{n'}(t_q) \times \left(c_{n(t_q)} - r_i\right)\right) \otimes \left(c_{n(t_q)} - r_i\right)}{\left|c_{n(t_q)} - r_i\right|^5}\right] \qquad \mathbf{G}(\mathbf{r}) = \sum_n x_n \mathbf{G}_n(\mathbf{r})$$

$$E_{n(r_i)} \approx \frac{\mu_0}{4\pi} * \frac{b_n - a_n}{2}\sum_{q=1}^{Q_n} w_q c_{n'}\frac{t_q}{\left|c_{n(t_q)} - r_i\right|} \qquad \mathbf{E}(\mathbf{r}) = \sum_n x_n \mathbf{E}_n(\mathbf{r})$$

$$L_{m,n} \approx \frac{\mu_0}{4\pi} * \frac{b_m - a_m}{2} * \frac{b_n - a_n}{2}\sum_{p=1}^{Q_m}\sum_{q=1}^{Q_n} w_p w_q \frac{c_{m'}(s_p) \cdot c_{n'}(t_q)}{\left|c_{n(t_q)} - c_{m(s_p)}\right|} \qquad \mathbf{W} = \sum_n \sum_m x_n x_m \mathbf{L}_{m,n}$$

Quadrature+Optimization Recap

$$\boldsymbol{x} \in \mathbb{R}^A \; : \text{describing design coefficients} \sim \boxed{\text{40 points}}$$

$$|\Omega_{\text{DSV}}| = N_G \; : \text{number of gradient eval points i.e. DSV points}$$

$$|\Omega_{\text{PNS}}| = N_E \; : \text{number of e-field eval points i.e. PNS points}$$

$$r_i^G \in \Omega_{\text{DSV}} \; : \; 1, ..., N_G \; \boxed{\text{DSV points} \sim 4k}$$

$$r_j^E \in \Omega_{\text{PNS}} \; : \; 1, ..., N_E \; \boxed{\text{PNS points} \sim 2.5k}$$

$$\boldsymbol{G} \in \mathbb{R}^{N_G \times A}, \quad \boldsymbol{G}_{i,n} = \boldsymbol{G}_n(r_i^G)$$

$$\boldsymbol{E} \in \mathbb{R}^{N_E \times A}, \quad \boldsymbol{E}_{j,n} = \boldsymbol{E}_n(r_j^E)$$

$$\boldsymbol{L} \in \mathbb{R}^{A \times A}$$

Note: Computing integrals requirement
SURFACE: ~40k vec-vec computation
MATRIX: ~Num_Loops*128 vec-vec computation

ADMM

We need a solver which takes in the matrices formed via quadrature and outputs the "optimal" coil parameters.

We would like the solver to:

- Relatively simple to implement
- **Good with large number of constraints**
- Ensures constraints are met!

ADMM is natural

- Splits up the problem into **multiple simpler problems**
- Robust proximal algorithm for inverse problems!

How does the algorithm work and how can we extend it to do surface optimization?

ADMM

Optimization Problem

$$\min_{\boldsymbol{\theta},\mathbf{x},G_{\min},L_{\max},E_{\max}} \; -\lambda_G G_{\min} + \lambda_L L_{\max} + \lambda_E E_{\max}$$

s.t.

$$\mathbf{e}_d^T \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x} \geq G_{\min}, \forall \mathbf{r} \in \Omega_{\text{dsv}}$$

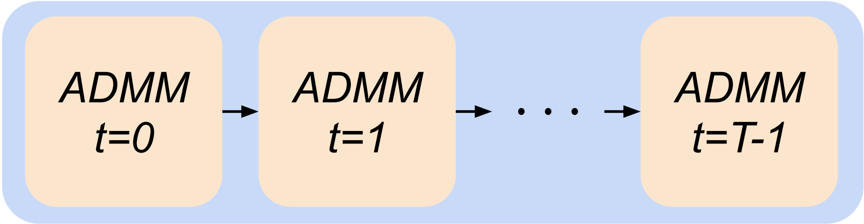
$$||\mathbf{E}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x}||_2 \leq E_{\max}, \forall \mathbf{r} \in \Omega_{\text{pns}}$$

$$\frac{1}{2} L_{\boldsymbol{\theta},n} x_n^2 \leq L_{\max}, \forall n \in \{1, \cdots, N\}.$$

Single ADMM Iteration

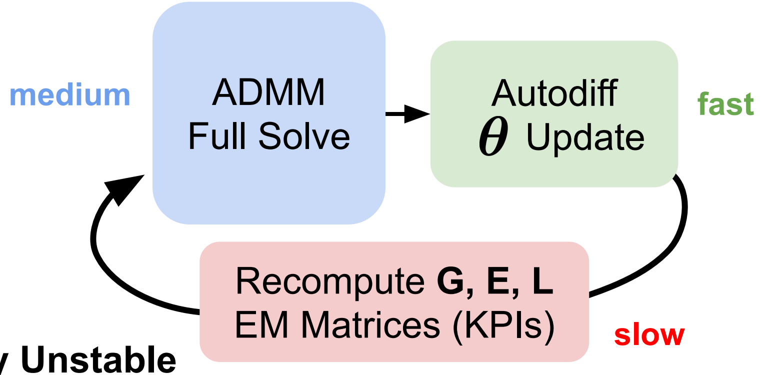
x-updates	Gmin Constraint	More Constraints
$\min_{\mathbf{x}} \|\|\mathbf{A}_t\mathbf{x} - \mathbf{b}_t\|\|_2$	$\min_{\mathbf{s},G} -\lambda G + \frac{\rho_G}{2}\|\|s - \mathbf{d}_G\|\|_2^2$ s.t. $\mathbf{s} \geq G$	$\dots$
$\mathbf{x}_{t+1} \leftarrow \mathbf{x}^\star$	$G_{\min,t+1} \leftarrow G^\star$, $\mathbf{s}_{t+1} \leftarrow \mathbf{s}^\star$	

Full Convex Solve



Stable and Robust

Unrolled Non-Convex Optimization



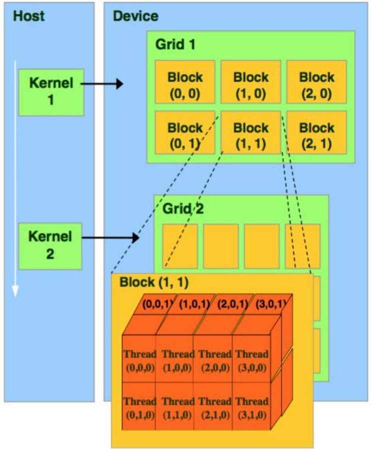
Very Unstable

Aside: GPU Optimization

Parallelization:

Notice: Each each loop can be processed independently and even each observation point r_i^G , r_j^E can be processed independently.

$$\mathbf{L}_{m,n} \approx \frac{\mu_0}{8\pi} \sum_{i=1}^{M} \sum_{j=1}^{N} \frac{\mathbf{J}_n^j \cdot \mathbf{J}_m^i}{\|\mathbf{r}_j - \mathbf{r}_i\|} \Delta S_i \Delta S_j$$



So we can partition in this way:
Let each block handle one (m,n) pair, thus each thread in the block handles some amount of (i, j) pairs.
Then we do a very efficient block sum operation called a **reduce**!

```
L = K.sum(dim=(-1, -2))          # reduce over quadrature indices p, q
W = 0.5 * (x[:, None] * L * x[None, :]).sum()
```

Implicit in Pytorch

Kernel Fusion:

Which runs faster?

```
# Unfused
dot = (cp_m[:, None, :, None, :] * cp_n[None, :, None, :, :]).sum(-1)
K = dot / dist
L = K.sum(dim=(-1, -2))

# Fused
L = torch.einsum('mpd,nqd,mnpq->mn', cp_m, cp_n, 1.0 / dist)
```

7x speedup with fused kernel

Kernel fusion is the process of combining many operations into one **if and only if** those intermediate operations are not to be used in the future.

The G and E matrix have a lot of overlapping computation so to reduce memory overhead we can combine their kernels!!

Outline

MRI & Gradients

Gradient Design and EM Modeling

Optimization & Computational Methods

$$\min_{\theta,\mathbf{x},G_{\min},L_{\max},E_{\max}} -\lambda_G G_{\min} + \lambda_L L_{\max} + \lambda_E E_{\max}$$

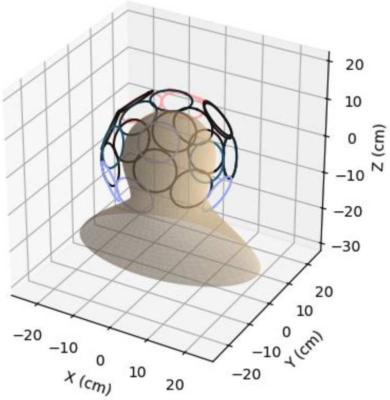
$$\text{s.t.}$$

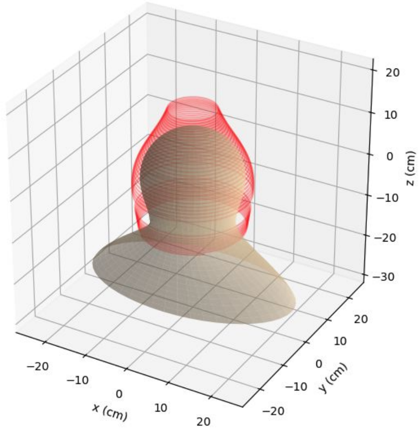
$$\mathbf{e}_d^T \mathbf{G}_\theta(\mathbf{r})\mathbf{x} \geq G_{\min}, \forall \mathbf{r} \in \Omega_{\text{dsv}}$$

$$||\mathbf{E}_\theta(\mathbf{r})\mathbf{x}||_2 \leq E_{\max}, \forall \mathbf{r} \in \Omega_{\text{pns}}$$

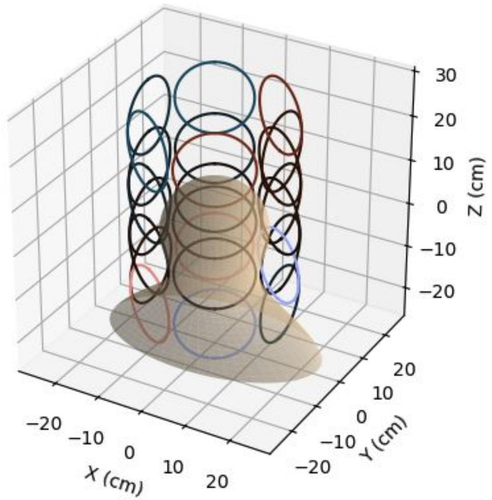
$$\frac{1}{2} L_{\theta,n} x_n^2 \leq L_{\max}, \forall n \in \{1, \cdots, N\}.$$

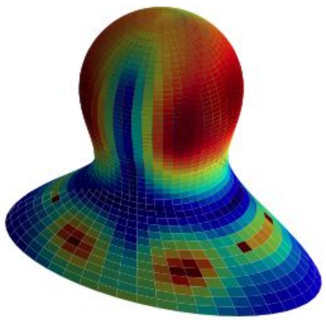
Simulation Results

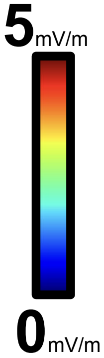


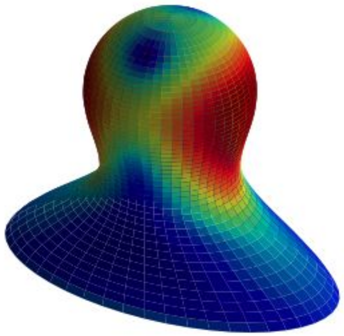


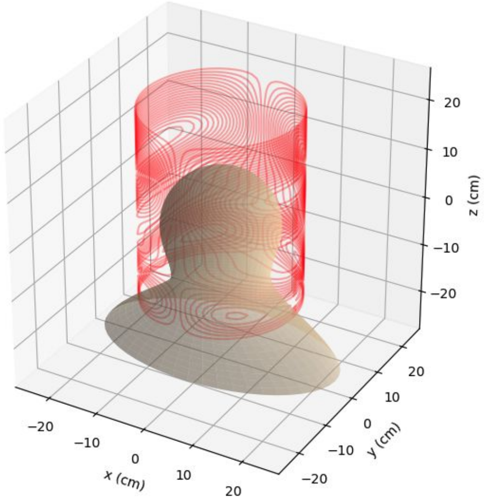
Matrix vs Stream Gradient: X-Axis *[5mV/m Emax, 300uH Lmax]*

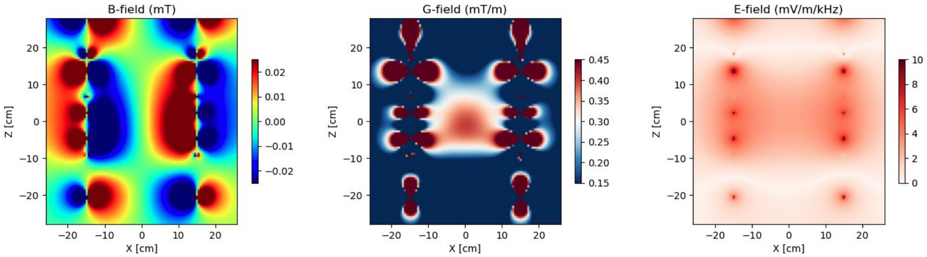


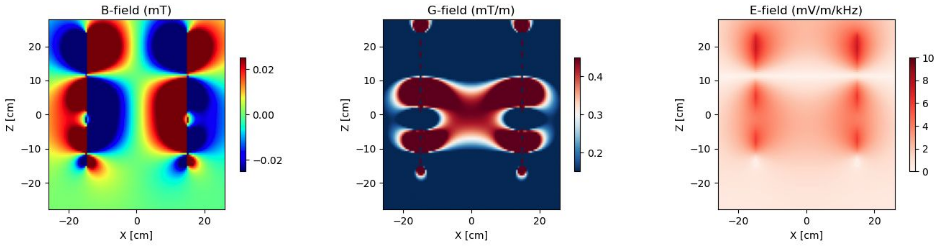








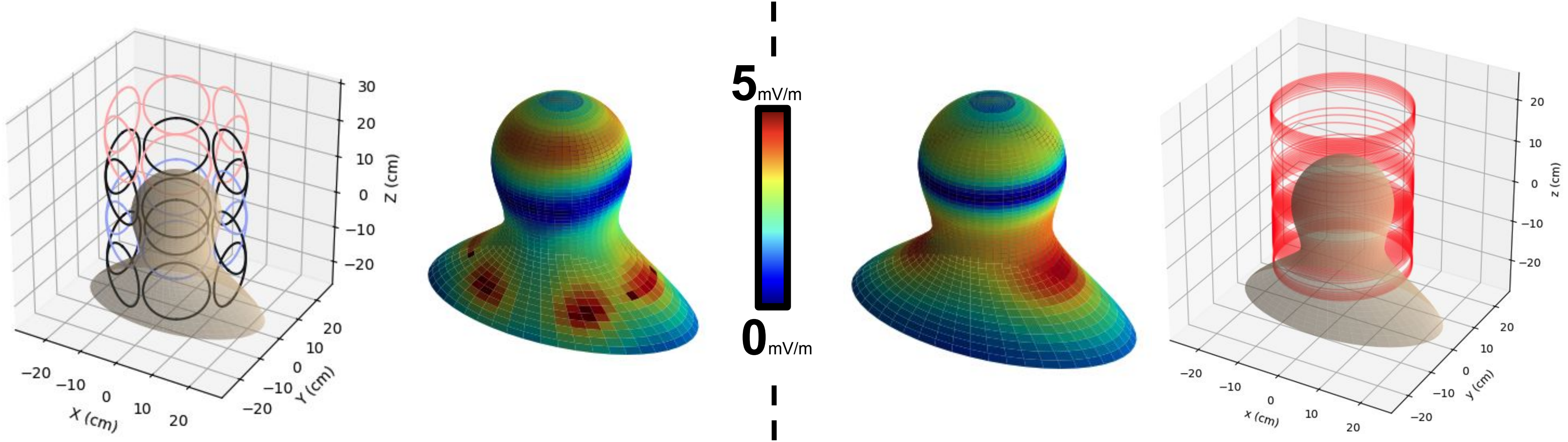


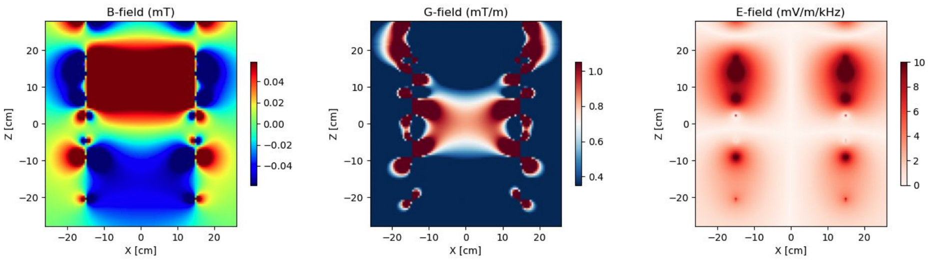


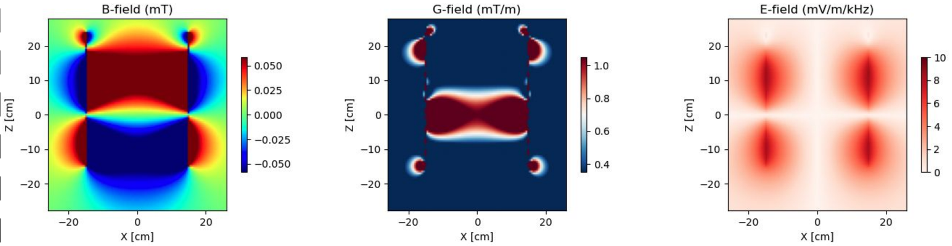
Gmin = **0.270**mT/m

Gmin = **0.274**mT/m

Matrix vs Stream Gradient: Z-Axis *[5mV/m Emax, 300uH Lmax]*



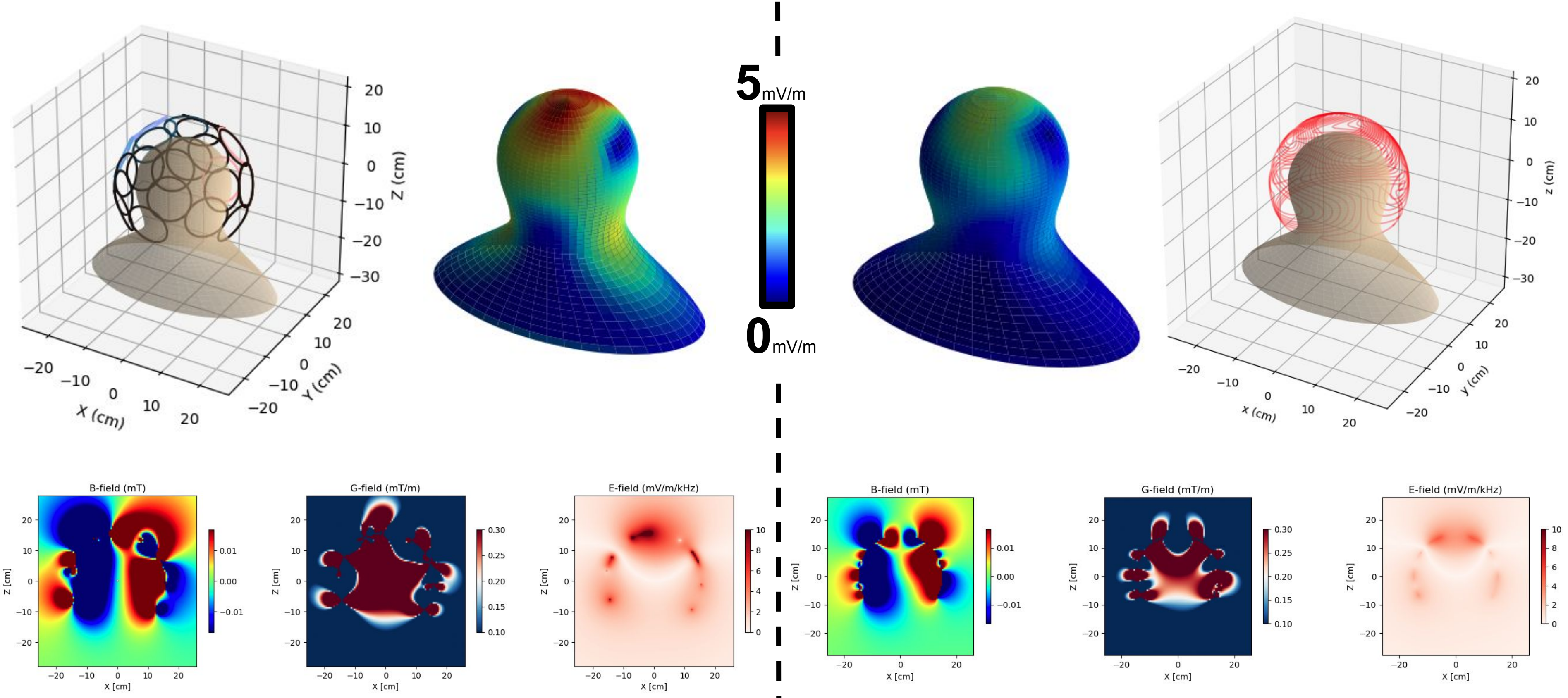




Gmin = **0.514**mT/m

Gmin = **0.448**mT/m

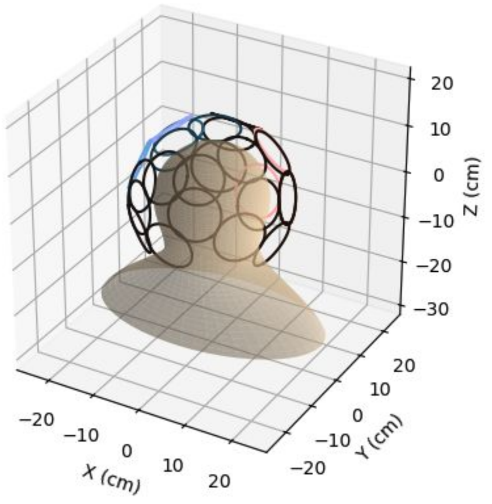
Matrix vs Stream Gradient: X-Axis *[5mV/m Emax, 300uH Lmax]*

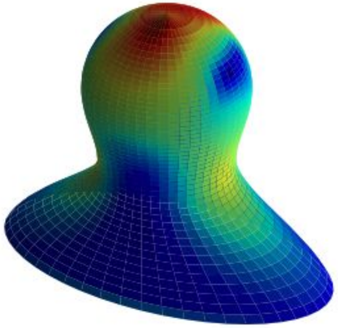


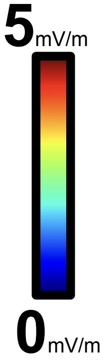
Gmin = **0.270**mT/m

Gmin = **0.192**mT/m

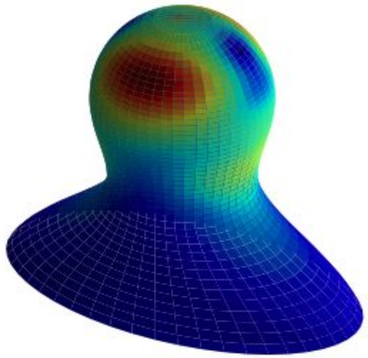
Matrix vs Stream Gradient: X-Axis *[5mV/m Emax, 300uH Lmax]*

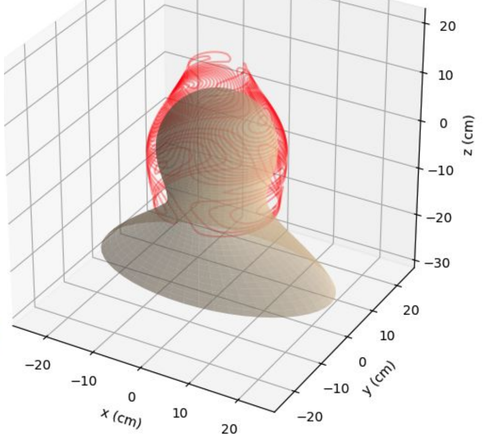


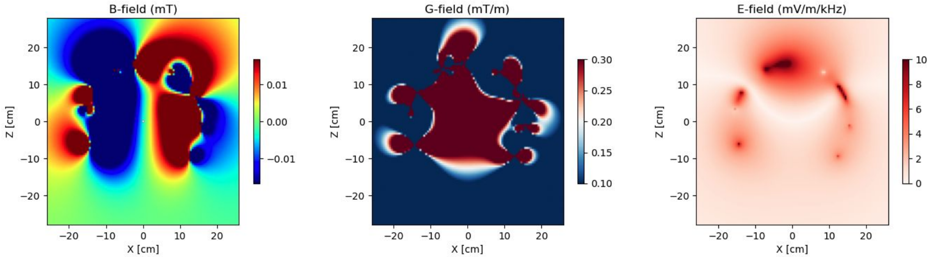




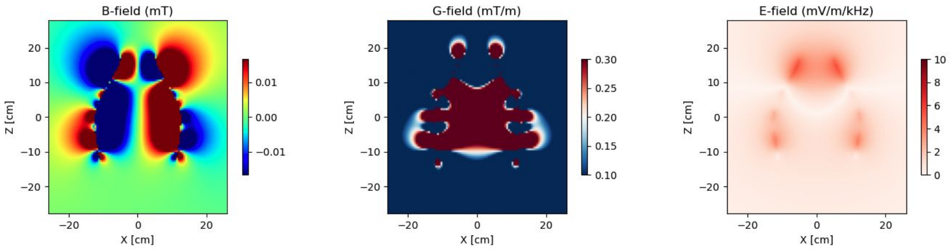
better surface





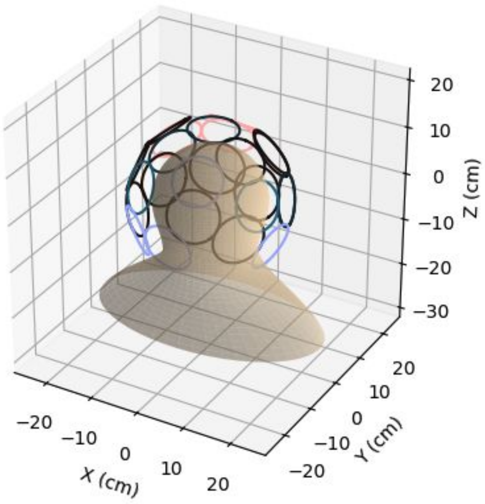


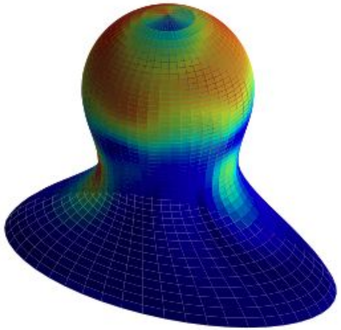
Gmin = **0.270**mT/m

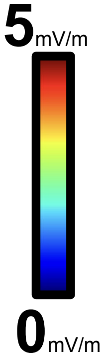


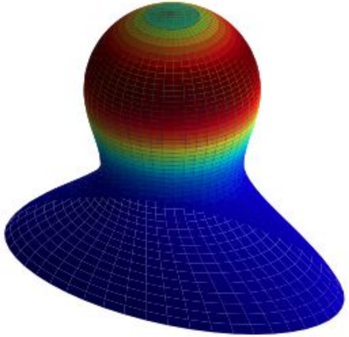
Gmin = **0.314**mT/m

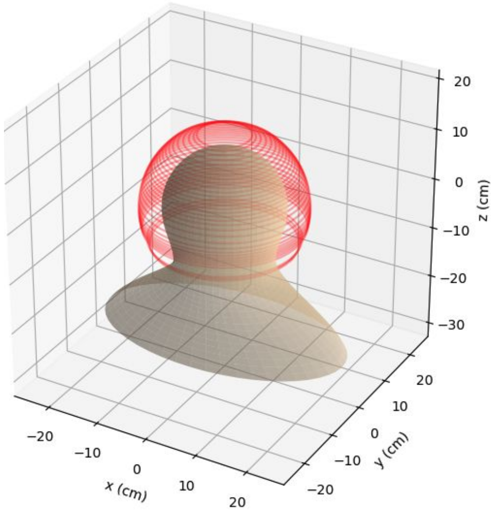
Matrix vs Stream Gradient: Z-Axis *[5mV/m Emax, 300uH Lmax]*

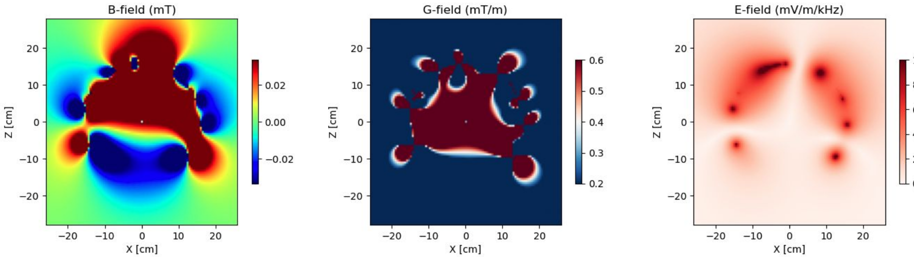


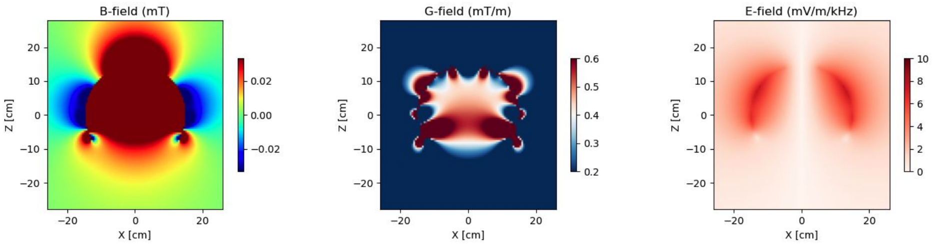










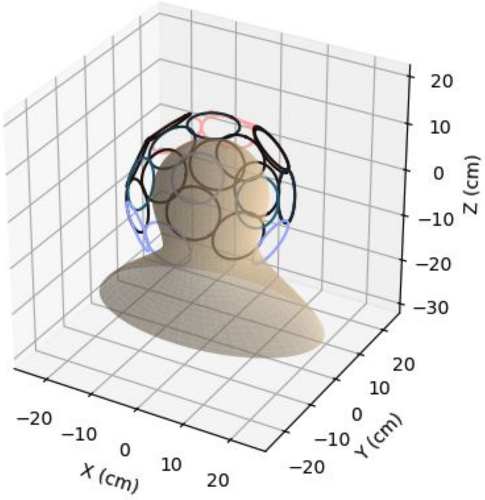


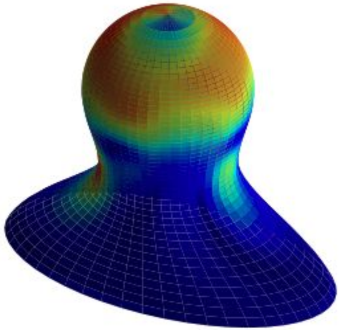
Gmin = **0.470**mT/m

Gmin = **0.427**mT/m

Matrix vs Stream Gradient: Z-Axis *[5mV/m Emax, 300uH Lmax]*

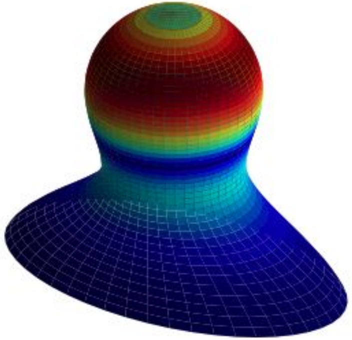
better surface

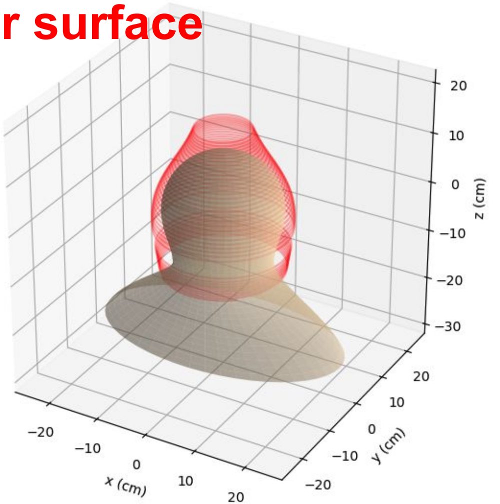


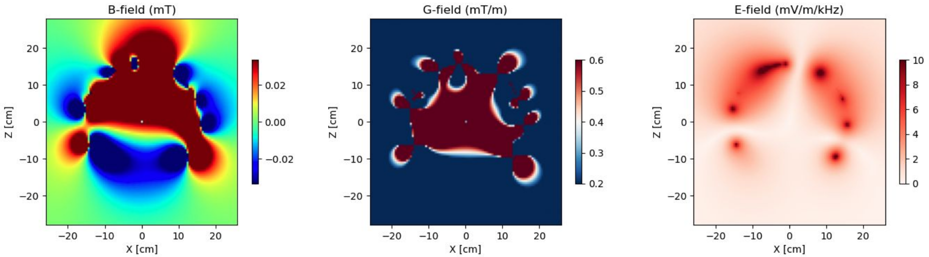


5mV/m

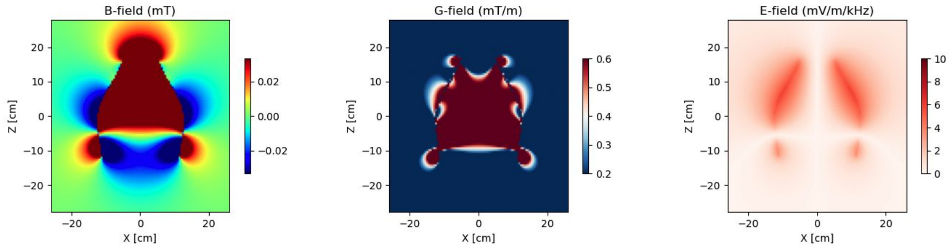
0mV/m







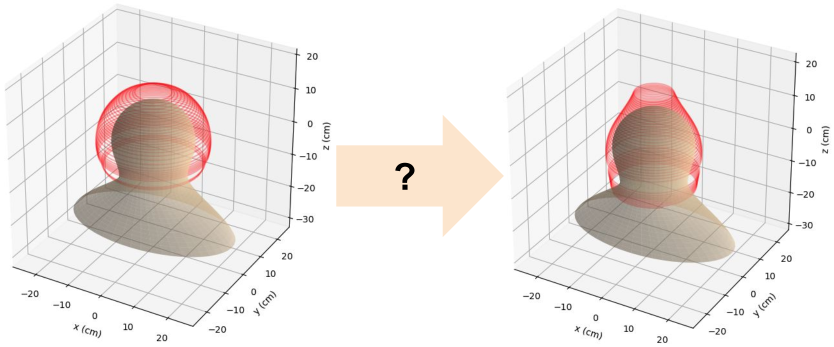
Gmin = **0.470**mT/m



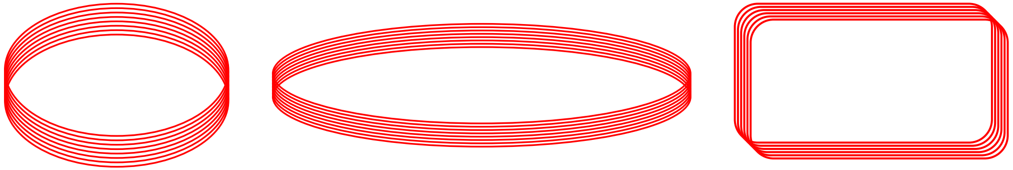
Gmin = **0.657**mT/m

To Be Determined

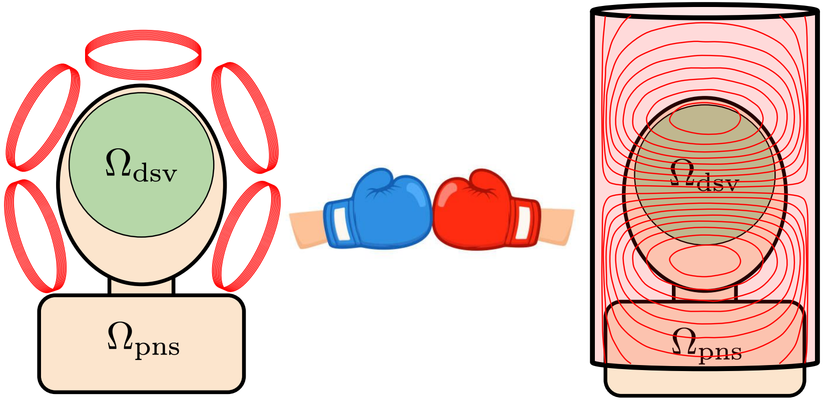
Can we reliably optimize surface geometry jointly with the surface current density?



What are optimal matrix coil shapes? Can these be optimized efficiently?



How do real world factors like amplifier cost, ease of use, and robustness influence the “right choice”?



EXTRA SLIDES

Optimization Formulation

Create a coil pattern which generates a strong gradient field at low power while mitigating PNS.

$\mathbf{x}$ => Parameters

$$\mathbf{G}(\mathbf{r}) = \nabla \mathbf{B}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{S'} \left[\frac{[\mathbf{J}(\mathbf{r})]_\times}{\|\mathbf{r}' - \mathbf{r}\|^3} - 3 \frac{(\mathbf{J}(\mathbf{r}) \times (\mathbf{r}' - \mathbf{r})) \otimes (\mathbf{r}' - \mathbf{r})}{\|\mathbf{r}' - \mathbf{r}\|^5} \right] dS'$$

=> Maximize

$$\mathbf{W} = \frac{\mu_0}{8\pi} \int_S \int_{S'} \frac{\mathbf{J}(\mathbf{r}') \cdot \mathbf{J}(\mathbf{r})}{\|\mathbf{r}' - \mathbf{r}\|} dS dS'$$

=> Minimize

$$\mathbf{E}(\mathbf{r}) = \frac{\mu_0}{4\pi} \int_{S'} \frac{\mathbf{J}(\mathbf{r}')}{\|\mathbf{r}' - \mathbf{r}\|} dS'$$

=> Minimize

Reminder: $\mathbf{J}(\mathbf{r}) dS \rightarrow I dl$

This is the equivalence of Matrix and Stream approaches.

Optimization Formulation

Create a coil pattern which generates a strong gradient field at low power while mitigating PNS.

$$\min_{\boldsymbol{\theta},\mathbf{x},G_{\min},L_{\max},E_{\max}} -\lambda_G G_{\min} + \lambda_L L_{\max} + \lambda_E E_{\max}$$

$$\text{s.t.}$$

$$\mathbf{e}_d^T \mathbf{G}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x} \geq G_{\min}, \forall \mathbf{r} \in \Omega_{\text{dsv}}$$

$$||\mathbf{E}_{\boldsymbol{\theta}}(\mathbf{r})\mathbf{x}||_2 \leq E_{\max}, \forall \mathbf{r} \in \Omega_{\text{pns}}$$

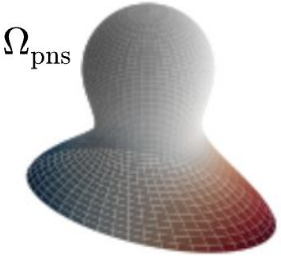
$$\frac{1}{2} L_{\boldsymbol{\theta},n} x_n^2 \leq L_{\max}, \forall n \in \{1, \cdots, N\}.$$

Continuous => Discrete

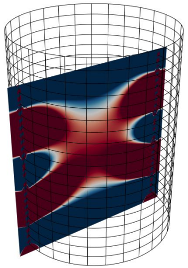
$$\forall \mathbf{r} \in \Omega_{\text{dsv}} \Rightarrow \forall \mathbf{r} \subset \Omega_{\text{dsv}}$$

$$\forall \mathbf{r} \in \Omega_{\text{pns}} \Rightarrow \forall \mathbf{r} \subset \Omega_{\text{pns}}$$

Ω_{pns}



Ω_{dsv}



Discrete Integral Computation - Quadrature

Convert $\int \rightarrow \sum$ and $dS \rightarrow \Delta S$

$$\mathbf{G}_n(\mathbf{r}_i) \approx \frac{\mu_0}{4\pi} \sum_{j=1}^{N} \left[\frac{[\mathbf{J}_n^j]_\times}{\|\mathbf{r}_j - \mathbf{r}_i\|^3} - 3\frac{(\mathbf{J}_n^j \times (\mathbf{r}_j - \mathbf{r}_i)) \otimes (\mathbf{r}_j - \mathbf{r}_i)}{\|\mathbf{r}_j - \mathbf{r}_i\|^5} \right] \Delta S_j$$

$$\mathbf{E}_n(\mathbf{r}_i) \approx \frac{\mu_0}{4\pi} \sum_{j=1}^{N} \frac{\mathbf{J}_n^j}{\|\mathbf{r}_j - \mathbf{r}_i\|} \Delta S_j$$

$$\mathbf{L}_{m,n} \approx \frac{\mu_0}{8\pi} \sum_{i=1}^{M} \sum_{j=1}^{N} \frac{\mathbf{J}_n^j \cdot \mathbf{J}_m^i}{\|\mathbf{r}_j - \mathbf{r}_i\|} \Delta S_i \Delta S_j$$

Panel based quadrature!

Left $\rightarrow \mathbf{J}^j = \mathbf{J}(\mathbf{r}_j)$, Right $\rightarrow \mathbf{J}^j = \mathbf{J}(\mathbf{r}_{j+1})$, Midpoint$\rightarrow \mathbf{J}^j = \mathbf{J}\left(\frac{\mathbf{r}_j + \mathbf{r}_{j+1}}{2}\right)$

Works for both Matrix and Stream approaches.
All of these integrals contain singularities defined by **close measurement and coil proximity**. Is this a problem?

Yes, but no!

Insert Midpoint quadrature graph

Discrete Integral Computation - Quadrature

Convert $\int \to \sum$ and $dS \to \Delta S$

$$\mathbf{G}_n(\mathbf{r}_i) \approx \frac{\mu_0}{4\pi}\sum_{j=1}^{N}\left[\frac{[\mathbf{J}_n^j]_\times}{\|\mathbf{r}_j-\mathbf{r}_i\|^3} - 3\frac{(\mathbf{J}_n^j \times (\mathbf{r}_j-\mathbf{r}_i)) \otimes (\mathbf{r}_j-\mathbf{r}_i)}{\|\mathbf{r}_j-\mathbf{r}_i\|^5}\right]\Delta S_j$$

$$\mathbf{E}_n(\mathbf{r}_i) \approx \frac{\mu_0}{4\pi}\sum_{j=1}^{N}\frac{\mathbf{J}_n^j}{\|\mathbf{r}_j-\mathbf{r}_i\|}\Delta S_j$$

$$\mathbf{L}_{m,n} \approx \frac{\mu_0}{8\pi}\sum_{i=1}^{M}\sum_{j=1}^{N}\frac{\mathbf{J}_n^j \cdot \mathbf{J}_m^i}{\|\mathbf{r}_j-\mathbf{r}_i\|}\Delta S_i \Delta S_j$$

Legendre quadrature!

Left $\to \mathbf{J}^j = \mathbf{J}(\mathbf{r}_j)$, Right $\to \mathbf{J}^j = \mathbf{J}(\mathbf{r}_{j+1})$, Midpoint$\to \mathbf{J}^j = \mathbf{J}\left(\frac{\mathbf{r}_j+\mathbf{r}_{j+1}}{2}\right)$

Works for both Matrix and Stream approaches.
All of these integrals contain singularities defined by **close measurement and coil proximity**. Is this a problem?

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Insert Midpoint quadrature graph